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Applicability of the coactivation method in assessing synergies of the scapular stabilizing muscles

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Hypothesis: This study investigated the feasibility of quantifying the coactivation levels in the evaluation of synergisms of the scapular stabilizing muscles during the elevation and lowering of the arms. This method, which quantifies the overlapping area of the normalized electromyographic (EMG) activity, could be applicable in assessing such muscular synergies.

Materials and methods: Both shoulders of 10 healthy volunteers were assessed. The coactivation of 3 synergic muscular groups (upper, middle and lower trapezius; upper trapezius and serratus anterior; middle trapezius and serratus anterior) and the isolated electromyographic activity of each muscle were analyzed.

Results: Analysis of variance revealed that the coactivation values between the synergic groups were different from the isolated levels of each respective muscle for both movements ($24.42 < F < 40.12$; df=6; $p < 0.001$). The coactivation values of all groups during elevation were different from those during the lowering of the arms ($13.31 < F < 959.92$; df=1; $p \leq 0.002$), with progressive increases in EMG activity during the elevation and decreases during the lowering of the arms.

Conclusion: The significant differences in EMG activity between the isolated and the coactivation methods indicated that the coactivation method was adequate to assess the scapular muscular synergies.

Level of evidence: Basic science study.

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Keywords: Electromyography; synergies; shoulder; scapular muscles

Muscular synergies have been defined as a group of muscles that are constrained to act as a single functional unit.[36] The important and essential roles of the synergic action of the muscles during the activities of daily living are well recognized: Muscular synergisms allow efficient motor control and coordination and also promote gains in motor skill learning, ability, and dexterity.[30,32,34,36,37] Several studies have already illustrated the roles of some synergic muscular groups on different joints of the body.[18,27,30,33] Among these joints, the shoulder girdle has been demonstrated as a classic example of the importance of muscular synergies because all muscles in this complex participate to some extent in all movements as synergic groups.[27]

The shoulder girdle shows the greatest amount of motion of the human body,[26,29] which is a result of the series of 4 joints that contribute to movement generation and the proximal attachment of the upper limbs at the small sternoclavicular joint.[29,39] The kinematics of these series of 4 joints must be well coordinated to allow the accomplishment of all functional activities performed by the upper limbs without impairing their different structures.[29] Among

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these 4 joints, the scapulothoracic plays an essential and fundamental role because it is responsible for the following functions:

- maintains the proper alignment between the glenoid fossa and the humeral head;
- provides a stable base for the glenohumeral muscles by maintaining optimal length-tension relationships during the movements of the upper limbs; and
- transfers the forces of the upper limbs from distal to the proximal regions of the body.[19]

All of these essential functions of the scapulothoracic joint are accomplished through the synergic activities of the scapulothoracic muscles.[26,27,39] The simultaneous and coordinated activation of these muscles, besides allowing the scapulae to perform their basic and essential roles, also generate and dissipate the forces occurring during the activity.[19]

Among the scapulothoracic muscles, the serratus anterior and the upper, middle, and lower trapezius muscles are considered the most important for the coordinated movement of the scapulothoracic joint.[21,28,29-31,34] Specifically, the upper trapezius and serratus anterior were pointed out as key synergic pairs for upward scapular rotation,[29-31,34] the middle trapezius and serratus anterior are important synergic pairs for maintaining the scapula against the thorax,[21,28,29] and finally, the 3 portions of the trapezius are considered essential for the coordination of the superior, inferior, and medial translation of the scapula during the elevation and lowering of the arms.[1,29,30] The lack of coordination between these muscles could generate abnormal scapular kinematics and lead to impairments of all of the important functions performed by the shoulder girdle.[30,31]

The isolated electromyographic (EMG) activity of the serratus anterior and the 3 portions of the trapezius has been described during the dynamic movement of elevating[1,21,22,25,40] and lowering of the arms[2,9,10] and provides important advances in the biomechanical knowledge of the shoulder complex. The results of these studies have been used to discuss the synergic actions of the scapular stabilizing muscles. The isolated activations of muscles do not provide a direct and objective measure of synergy, however, because muscular synergy refers to a group of muscles that is constrained to act as a single functional unit.[36] This fact limits the complete understanding of how the muscles effectively contribute to movement generation and to dynamic stabilization.[32,36,37]

It is essential to investigate an applicable method that provides a way to directly and objectively measure muscular synergic actions, principally at the shoulder complex, where dynamic stability is essential to accomplish all daily functional activities of the upper limbs[26,29,30,39] and where all muscles participate to some extent in all movements as synergic groups.[27] One simple and promising way to measure simultaneous muscular activation and to obtain direct and objective information about their synergism is to use methodologies applied to quantify the levels of coactivation or cocontraction of pairs of muscles. Among the proposed methods to measure simultaneous muscular activation, the one presented by Unnithan et al[38] quantifies the overlapping area of the normalized EMG signal and seems to be the most advantageous.[12,38] This method allows the quantification of the coactivation at any point in time and also provides a mean value of the intensity of the coactivation during a specific moment. Moreover, this method does not demand muscular classifications of agonists and antagonists, as is usually reported in studies of cocontraction,[12,38] which extends its applicability to assess the synergic muscular activation.

Therefore, the aim of this study was to evaluate the method proposed by Unnithan et al[38] to assess the synergies of the scapular stabilizing muscles during the elevation and lowering of the arms. The synergies of the following scapulothoracic groups were evaluated: the upper, middle and lower trapezius; the upper trapezius and serratus anterior; and the middle trapezius and serratus anterior. To allow a better investigation of this method, isolated EMG activity level of each muscle was also evaluated.

Materials and methods

Both shoulders of 10 healthy volunteers without shoulder complaints or histories of previous shoulder injuries were assessed, totalling 20 evaluated shoulders. Volunteers were excluded if they participated in physical activities that demanded overhead tasks, such as volleyball and handball, even though they could not have symptoms of shoulder impairments, because they might demonstrate specific characteristics of kinematic[21] and synergic patterns that might interfere with the results. Individuals who were unable to complete the tests were also excluded.

Surface EMG (model MP100WSW, Biopac Systems Inc, Santa Barbara, CA) was used to determine bilateral EMG activity of the serratus anterior and the upper, middle, and lower trapezius muscles. This device has 2 amplifiers connected to a microcomputer that has an input impedance of 2 MΩ, a common-mode rejection ratio (CMRR) of 1000 MΩ, and allows data acquisition at frequencies from 10 to 1000 Hz. There were used eight pairs of active surface electrodes (TSD 150) with a diameter of 11.4 mm, a distance between electrodes of 20 mm, an impedance of 100 MΩ, and a reference electrode. The 3 Pro isokinetic dynamometer (Biodex Medical Systems Inc, Shirley, NY) was used to stabilize individuals during the performance of maximal voluntary contractions (MVC), which were used to normalize EMG activity. This method has been recommended and frequently used to normalize the EMG activity due to its reproducibility and ability to distinguish changes in muscular activation in response to different force magnitudes.[3,6,20,35]

To analyze the EMG activity of the scapular stabilizing muscles in the scapular plane, defined as 30° anterior to the frontal plane,[29,40] a home-made device was used[10,25,40] to standardize and assure the positioning of the upper extremities. The scapular plane has been recommended and is widely used for the assessment of upper limb movements[10,21,25,40] because it is central to most daily

and occupational activities and provides the best mechanical advantages for the performance of the glenohumeral muscles.[29] The device was divided into increments of 5°, from 0° to 180°, and was adjusted to the individuals' height and had a support base capable of maintaining equal distances and levels between the lower extremities. A digital video camera with a light signal connected to both the lens and 1 channel of the EMG was used to register all movements and to synchronize the images and the signals of the EMG activity.

Before the data were collected, the volunteers were informed about the study objectives and asked to sign a consent form that was previously approved by the University Ethics Review Board. All participants were evaluated by the same investigator to collect demographic data and any other relevant information.

The skin was then shaved and rubbed with alcohol to reduce impedance, and surface active electrodes were placed in pairs and parallel to the muscle fibers, following the recommendations of Cram et al,[6] and the reference electrode was placed over the right acromion.[6,10,25] The quality of the signal for each muscle was assured by asking participants to perform a manually resisted contraction.[6] To perform the MVC, subjects were seated on a dynamometer chair, performed the movements in a closed kinetic chain,[10,13] and received standardized verbal encouragement. No warm-up was performed, and the EMG activity related to the MVC of each muscle was recorded 3 times with the duration of each contraction of about 6 seconds.[6]

Barefoot participants were then instructed to assume a standing position on the device, which was adjusted according their height, and were asked to raise and lower their arms at a comfortable, self-selected speed following a verbal command. Simultaneously with this command, the light connected to the lens of the camera and to one of the EMG channels was activated, and its signal was used for synchronization of both the images and EMG data. The participants performed bilateral elevation and lowering of the arms with the palmar surfaces of the hands maintaining contact with the surface of the device. They were also asked to focus on an anterior eye-level target while they performed 6 trials. The data from the 6 trials were recorded and stored for analysis, and information related to the 3 trials that showed the closest speed values was selected for analyses.

The selected files of the image and of the EMG were first synchronized using a simple calculation considering the processing frequencies of all of the software. The onset of the light was recorded with the image and with the EMG signal and was processed by VirtualDub 1.6.10 software (GNU General Public License) at a frequency of 33Hz and by the Acqknowledge software (Biopac) at a frequency of 1000Hz. It was possible to synchronize the data by considering the moment of the onset of the light in each file and the processing frequency for each program. The precision of this method was investigated, and the error rate was only 0.1%.[10]

After the synchronization process, the beginning and end of the elevation and lowering of the arms were determined. In addition, for the elevation, the beginning and end of the following phases were determined: from the beginning to 30° (phase 1a); 30° to 60° (phase 2a); 60° to 90° (phase 3a); 90° to 120° (phase 4a); 120° to 150° (phase 5a); and 150° to the end (phase 6a). For the lowering of the arms, the same phases were considered: from the beginning to 150° (phase 1b); 150° to 120° (phase 2b); 120° to 90° (phase 3b); 90° to 60° (phase 4b); 60° to 30° (phase 5b); and 30° to the end (phase 6b). This procedure was completed by focusing on the image of the third finger for both sides moving along the surface of the device. Once the frequency of the image was captured, it was possible to conduct precise frame-by-frame analyses at intervals of 33 milliseconds of the finger position.

Signals were processed with specific routines developed with MatLab 7.0.0.19920 software (MathWorks Inc, Natick, MA) and were full-wave rectified and filtered, following previously described procedures.[10,11,25] Root mean squares were used to quantify the EMG activity of the elevation and lowering of the arms as well as during the intervals of each phase. Two seconds of highest activity of the MVC of each muscle were selected to normalize the EMG activity.[10,11] For the isolated EMG activity, the parameters selected for analyses were the amounts of EMG activity of each muscle for each complete movement as well as for each phase, expressed as percentages of the MVC. For the coactivation, the parameters were the amount of coactivation of the 3 portions of the trapezius (upper, middle and inferior), the upper trapezius and serratus anterior, and the middle trapezius and serratus anterior. The amount of coactivation was calculated as the overlapping areas of the normalized EMG activity curve of each synergic group of muscles, as previously described and used.[21,40] This overlapped area represented the intensity of the simultaneous muscular activation or coactivation levels[11,12] of the assessed synergic groups. Once again, specific procedures developed at MatLab were used to obtain the coactivation levels.

Descriptive statistics and tests for normality and homogeneity of variance were calculated for all outcome variables by using SPSS 13.0 software (SPSS Inc, Chicago, IL). Repeated measures analysis of variance (ANOVA), followed by planned contrasts, were used to compare the amount of isolated EMG activity with the amount of coactivation for all assessed muscles, with a significance level of $\alpha < 0.05$.

Results

The 10 study volunteers (6 women and 4 men) had a mean age of 29.0 ± 5.4 years (range, 21-36 years), had a mean body mass of 58.2 ± 11.1 kg (range, 47.0-81.5 kg), and were a mean height of 1.7 ± 0.1 m (range, 1.6-1.8 m). Eight were sedentary and right-side—dominant, and none performed activities that demanded overhead movements. The means of movement excursion, time, and velocity during the elevation and lowering of the arms were 164.5° ± 2.8° and 160.5° ± 7.7°, 3.5 ± 1.2 and 3.7 ± 1.3 seconds, and 50.6° ± 11.3° and 47.1° ± 11.6° per second, respectively. No significant differences were found in movement velocities between elevation and lowering of the arms ($P = .14$).

Figure 1 demonstrates an example of the overlapping areas of the normalized EMG activity for each evaluated synergic group. These areas represent the intensity of simultaneous muscular activation or coactivation levels of the upper, middle, and lower trapezius (Figure 1, *A*); upper trapezius and serratus anterior (Figure 1, *B*); and middle trapezius and serratus anterior (Figure 1, *C*).

Synergic activity of the upper, middle, and lower trapezius

Figure 2 shows the means of the isolated EMG and the coactivation EMG activity of the upper, middle, and lower

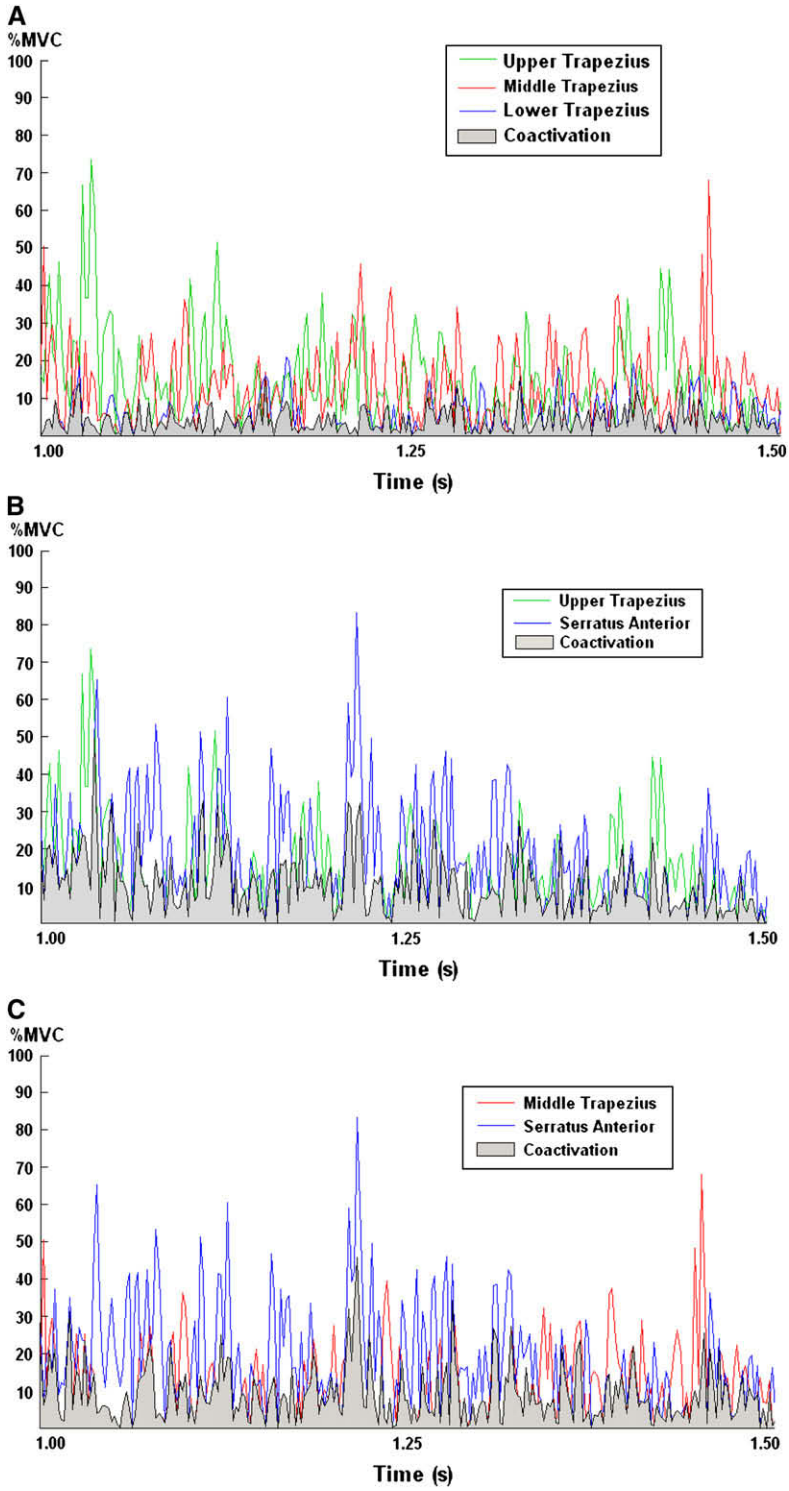

Figure 1 Overlapping areas of the normalized electromyographic activity for each synergic muscle group: **(A)** upper, middle and lower trapezius, **(B)** upper trapezius and serratus anterior, and **(C)** middle trapezius and serratus anterior. *MVC*, maximal voluntary contractions.

trapezius during the investigated phases of elevation and lowering of the arms. From the beginning of the elevation to the end of the lowering, the isolated EMG activity of the upper trapezius ranged from 18.65% ± 9.72% to 4.46% ± 2.92%, of the middle trapezius from 14.16% ± 8.37% to 8.23% ± 5.73%; and of the lower trapezius from 17.55% ± 10.65% to 12.29% ± 9.65%. Their coactivation levels ranged from 3.84% ± 1.54% to 1.42% ± 0.69%.

The lower and higher levels of the isolated EMG activities of the upper, middle, and lower trapezius were

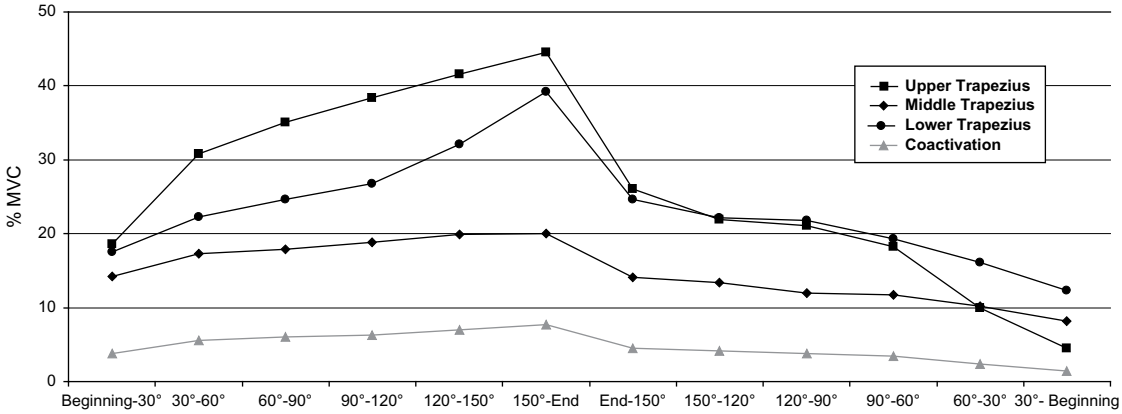

Figure 2 Isolated electromyographic activity (percentage maximal voluntary contractions [*%MVC*]) of the upper, middle, and lower trapezius, and coactivation levels among the upper, middle, and lower trapezius in the different phases of elevation and lowering of the arms.

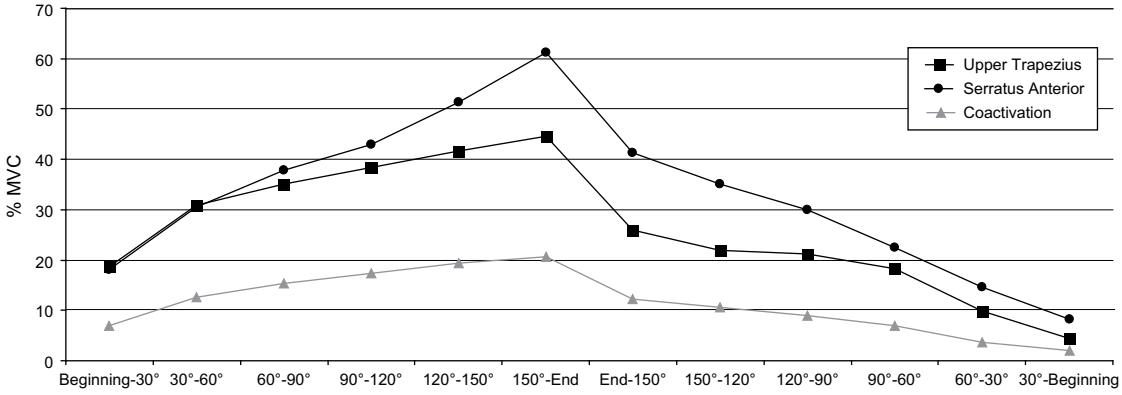

Figure 3 Isolated electromyographic activity (percentage maximal voluntary contractions [*%MVC*]) of the upper trapezius and serratus anterior, and coactivation levels between the upper trapezius and serratus anterior in the different phases of elevation and lowering of the arms

4.46% to 44.57%, 8.23% to 20.04%, and 12.29% to 39.23%, respectively. The lower and higher levels of the coactivation were 1.42% and 7.74%.

Synergic activity of the upper trapezius and serratus anterior

Figure 3 demonstrates the means of the isolated EMG and the coactivation EMG activity between the upper trapezius and serratus anterior during the phases of elevation and lowering of the arms. From the beginning of the elevation to the end of the lowering, the isolated EMG activity of the upper trapezius ranged from 18.65% ± 9.72% to 4.46% ± 2.92% and of the serratus anterior from 18.08% ± 7.21% to 8.18% ± 4.71%. Their coactivation levels ranged from 5.70% ± 2.22% to 2.21% ± 1.21%.

The lower and higher levels of the isolated EMG activities of the upper trapezius and serratus anterior were 4.46% and 44.57%, and 8.18% and 61.21%, respectively. The lower and higher levels of coactivation were 2.06% and 20.73%.

Synergic activity of the middle trapezius and serratus anterior

Figure 4 provides the means of the isolated EMG and the coactivation EMG activity between the middle trapezius and serratus anterior during the various phases of elevation and lowering of the arms. From the beginning of the elevation to the end of the lowering, the isolated EMG activity of the middle trapezius ranged from 14.16% ± 8.37% to 8.23% ± 5.73%; and of the serratus anterior from 18.08% ± 7.21% to 8.18% ± 4.71%. Their coactivation levels ranged from 7.02% ± 2.50% to 2.06% ± 2.43%.

The lower and higher levels of the isolated EMG activities of the middle trapezius and serratus anterior were 8.23% and 20.04%, and 8.18% and 61.21%, respectively. The lower and higher levels of coactivation were 2.21% and 11.54%.

Comparisons between isolated and coactivation activity levels

The means of the isolated EMG activity of all investigated muscles are provided in Figure 5, *A*. The means of the

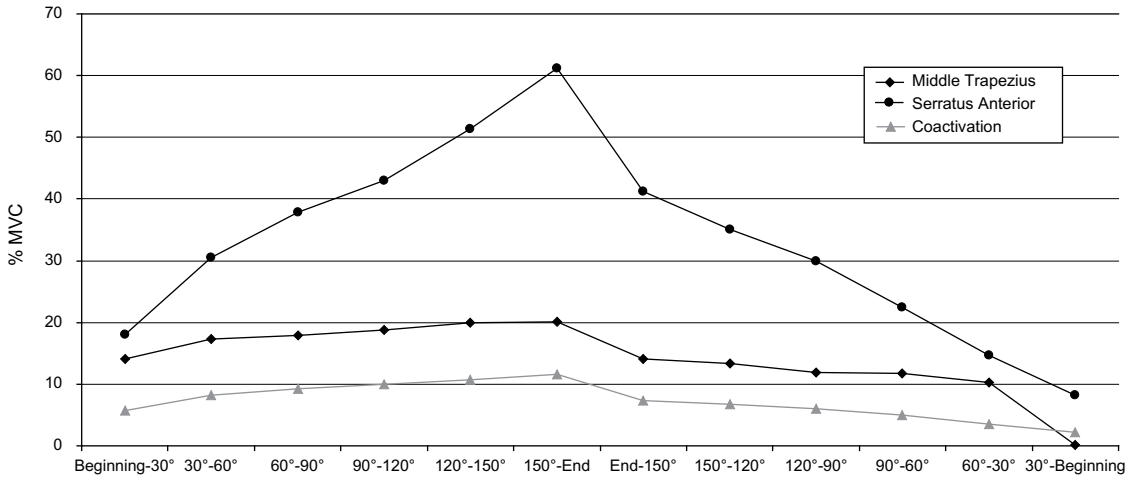

Figure 4 **(A)** Isolated electromyographic activity and **(B)** coactivation levels during the complete movements of elevation and lowering of the arms. *MVC,* Maximal voluntary contractions.

coactivation levels of all investigated synergic groups, considering the complete movement of elevation and lowering of the arms, are provided in Figure 5, *B*. For the elevation, the means of the isolated EMG activities were 44.61% ± 14.05% for the serratus anterior, and 6.93% ± 14.99% for the upper, 18.51% ± 10.00% for the middle, and 29.45% ± 17.57% for the lower trapezius. On the other hand, the means of the coactivation EMG activities were 21.63% ± 10.45% for the upper, middle and lower trapezius group; 51.12% ± 13.58% for the upper trapezius and serratus anterior group; and 33.45% ± 18.10% for the middle trapezius and serratus anterior group. For the lowering of the arms, the means of the isolated EMG activities of were 27.44% ± 10.21% for the serratus anterior and 18.43% ± 8.00% for the upper, 11.89% ± 7.00% for the middle, and 19.82% ± 16.98% for the lower trapezius. The means of the coactivation EMG activities were 3.14% ± 1.31% for the upper, middle and lower trapezius group; 6.86% ± 1.78% for the upper trapezius and serratus anterior group; and 4.89% ± 1.92% for the middle trapezius and serratus anterior group.

The ANOVA demonstrated significant differences between the amount of isolated EMG and the coactivation EMG levels during the elevation ($F = 24.42$; df=6; $p < 0.001$) and the lowering of the arms ($F = 40.12$; df=6; $p < 0.001$). In addition, all 3 investigated synergic pairs showed significant differences in EMG activity from each other during both the elevation ($23.61 < F < 134.43$; df=1; $p < 0.001$) and lowering of the arms ($13.31 < F < 959.92$; df=1; $p \leq 0.002$).

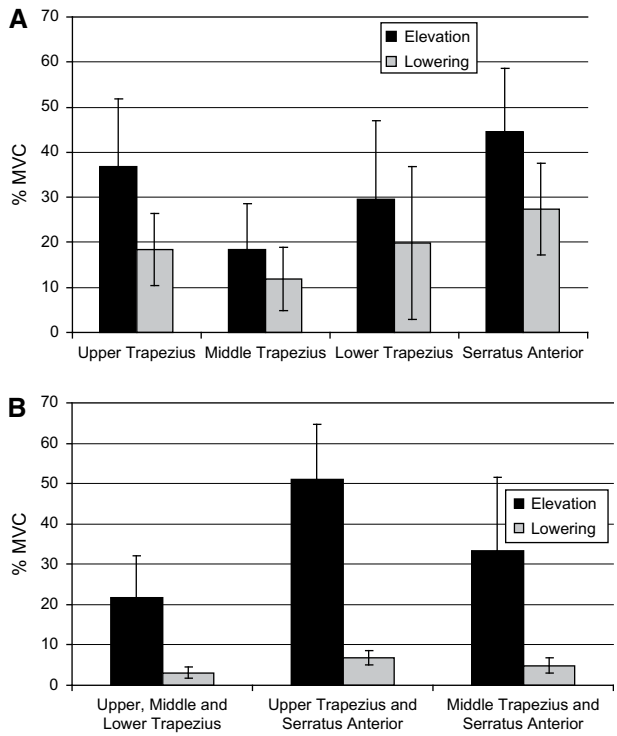

Figure 5 **(A)** Isolated electromyographic activity of all investigated muscles and **(B)** coactivation levels of the synergic groups during the movements of elevation and lowering of the arms. The *error bars* designate the standard deviation. *MVC,* Maximal voluntary contractions.

Discussion

The aim of the present study was to evaluate the method outlined by Unnithan et al[38] for the assessment of the muscular synergies of the scapular stabilizing muscles during the elevation and lowering of the arms. These findings demonstrated that the coactivation levels between each investigated synergic muscle group were different from those of the isolated EMG activation of the respective muscles, both for the elevation and lowering of the arms. In addition, the coactivation levels of each synergic group were also significantly different during both the elevation and lowering of the arms. The shapes of the curves in Figures 2, 3, and 4 demonstrated coactivation patterns coherent with what was expected: progressive increases in

EMG activity during the elevation of the arms with progressive decreases during their lowering. These results associated with the theoretic statements related to the synergic actions of the scapular stabilizing muscles pointed out the applicability of this method in assessing the scapular stabilizing synergies during both movements.

Traditionally, studies have reported the isolated EMG activity levels for different muscles in various situations[1,6,21,22] and their findings were important for a better understanding of the muscular behavior during human movements performed by healthy individuals and those with various impairments. During functional activities, however, the muscles are not contracted in isolation but in synergies, which allow the efficient coordination of the motor apparatus.[30,32,34,36,37] Despite being a common way to study muscular synergy, the isolated EMG activity does not provide a direct and objective measure of synergies. The results of the present study supported this evidence, because the values obtained with the isolated EMG activation were significantly different from those obtained with the coactivation method.

A reasonable explanation for the lower coactivation levels compared with the isolated EMG level was that the method used to obtain the coactivation levels analyzed the overlapping areas of the normalized isolated EMG activity curve.[21,40] This overlapping area represented the intensity of the simultaneous muscular activation or coactivation levels[11,12] of the assessed synergic groups. Therefore, specific muscular activations that were not simultaneous to the activation of the other muscle(s) of the synergic group were not considered as coactivation activities. Muscles can be activated for different purposes: to perform the main or secondary movements, to stabilize a structure that can be directly or indirectly related to the desired movement, or to dissipate forces generated by other muscles. Within the investigated method in the present study, the activation of interest was the simultaneous one, which is related to the muscular synergies of the group of muscles that are constrained to act as a single functional unit[36] and is considered sufficient to describe muscular functional activity.[24] In a previous study that investigated the coactivation levels of the vastus lateralis and biceps femoris muscles during gait, the reported mean coactivation values were also lower than the isolated ones,[11] as was found in the present study.

To be able to control such a complex system as in the shoulder girdle, muscles must be simultaneously and synchronously activated to produce movements and to stabilize the joint.[36,37] Thus, the muscular synergisms may be related to the necessary adjustments to counteract a set of resultant forces produced during the movements, and it is possible that alterations in these synergies would lead to dysfunctions.[36,37] From these assumptions it can be concluded that for a better understanding of the movements and their dysfunctions, as well as of the processes underlying gains in motor skill learning, ability and dexterity,[32] the assessment of specific variables more related to muscular synergies should be included.[36,37]

The study of muscular synergic behavior is even more important for the shoulder complex, where all structures, muscles, and joints must work in perfect harmony to permit all possible movements in an accurate and efficient way.[19] The shoulder functions need to meet a balance between mobility and stability that require fine-tuned muscle coordination.[39] Specifically, the simultaneous activation of the scapulothoracic muscles is considered important to maintain an adequate scapular position during all movements and then contribute to the stability of the shoulder complex.[31,41] Therefore, studies based on isolated muscular activation can only provide limited information. However, the data related to the synergic actions of these muscle groups could provide a better understanding of the role of one muscle in relation to the other or how a specific muscle acts in synergy with others,[12] adding more information to advance the understanding of the synergic action of those muscles, which would allow better comprehension of the functional performance of the shoulder complex and of all the common dysfunctions that could compromise normal joint functions and lead to disabilities.[7,15,17,19,30,37]

In the shoulder complex, all muscles participate to some extent in all movements as synergic pairs, and the scapulohumeral muscles, in particular, contract as a group.[27] Different synergic groups among the scapulothoracic muscles have been identified and are constituted according to the movements of the arm and the required demands for stabilization. The simultaneous and coordinated actions between the upper trapezius and the serratus anterior muscles have been pointed out as crucial for upward scapular rotation.[34] Anther important synergic group identified in the scapulothoracic joint is the middle trapezius and serratus anterior, responsible for maintaining the scapula against the thorax.[36,37] The lack of coordination between these pairs of muscles could generate abnormal scapular kinematics and lead to impairments of all the important functions performed by these anatomic structures. Studies that quantified the isolated activity of the scapular stabilizing muscles in individuals with impingement syndrome found altered scapular muscular recruitment patterns[4] and strength,[5] impaired shoulder kinematics, such as scapular tipping and winging, and imbalances of the activity patterns of the serratus anterior and the lower and upper trapezius.[21,23] The use of a method that quantifies the coactivation levels between these synergic groups could provide a better comprehension of the relationships between altered kinematics and muscle weakness and imbalances in common shoulder dysfunctions.

Previous studies that applied any methods related to the assessment of muscular coactivation levels aimed at investigating the actions between 2 opposing muscles for the analysis of dynamic joint stability,[11] motor coordination,[8] stages of motor learning,[14] or for the description of altered movement patterns.[38] In the present study, the same method was used to provide a direct and objective measure to assess the muscular synergisms between the scapular stabilizing muscles during the dynamic movements of

elevation and lowering of the arms. This represents completely different applications of the method and interpretation of the outcomes related to the coactivation levels. In addition, the method was investigated only for the synergic groups of the scapulothoracic joint. Future studies are necessary to investigate the applicability of this method in the evaluation of other synergic groups of other joints.

Finally, future studies should consider other potential synergies that may also be important in the shoulder girdle, such as the lower trapezius and serratus anterior to control the scapular external rotation and posterior tilt,[1] the lower trapezius and latissimus dorsi for arm elevation and scapular stabilization, the pectoralis minor and middle trapezius to stabilize scapular rotation,[16] as well as the rotator cuff group to stabilize the shoulder joint.[16,30] The synergic action of these muscles during specific tasks performed by both healthy and impaired volunteers should also be investigated because there would be task-specific synergies that could be effective.[36,37] The results of these investigations will be helpful for clinical assessments and interventions and allow rehabilitation strategies to be directly applied to specific populations and tasks.

Conclusions

The significant differences in EMG activity levels obtained with the isolated and coactivation methods indicated that the coactivation method may be selected to study muscular synergies. These results associated with the theoretical statements related to the synergic actions of the scapular stabilizing muscles, pointed out that the coactivation method was adequate to assess the synergies between the scapular muscles during the elevation and lowering of the arms. Therefore, this method may be applied not only to describe the coactivation levels between two opposing muscles, but also to provide direct and objective measures of the muscular synergies during dynamic movements.

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